

Week 12: Safe Book Class

Overview

This assignment turns Week 12 concepts into a small, finishable Java program or project milestone. Start from the provided template, preserve the public class filename, and run the program in IntelliJ before submitting.

Objectives

- Design private fields and a constructor.
- Throw an `IllegalArgumentException` for invalid data.
- Catch and explain an exception.

Required reading - one new chapter

- [Think Java Chapter 11 - Designing Classes](#)

Optional supplemental reading - previously covered chapters

- [Chapter 5 - Conditionals and Logic](#)

Interactive practice

- [Runestone: writing classes exercises](#)

Supplemental videos

- [OOP](#)
- [Constructors](#)

Complete these steps

- Create `Week12BookClass` and a class named `BookDemo`.
- Copy the starter into `BookDemo.java`.
- Keep `title` and `pages` private.
- In the constructor, throw new `IllegalArgumentException` when `title` is blank or `pages` is not positive.
- Initialize both fields only after validation.
- Complete `getTitle`, `getPages`, and `describe`.
- In `main`, create and print `Charlotte's Web` with 192 pages.
- Use a `try/catch` block to attempt one invalid `Book` and print the exception message.
- Run and confirm the valid description and validation message appear.
- Submit `BookDemo.java`.

What will be checked

Use this list as your final review before submitting.

- `Book` has private `title` and `pages` fields.
- The constructor throws `IllegalArgumentException` for invalid data.
- `main` uses `try/catch`.
- The valid book and an understandable validation message are printed.
- The submitted file contains no unfinished `TODO` marker from the starter template.